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Chemistry Mock

Answers and Marking Scheme

M: method mark A: answer mark (only if the related M mark is awarded) B: independent mark C: communication mark

Underlined words are key words that must appear in the answer. Chemical equations must be balanced; contradictory answers score no marks for that part. For revision only; other valid answers should also be credited.

Section A Industrial Chemistry

1. (20 marks)

(a)(i1) (1)

B1 Method 1: 8 ÷ 52 = 15.4%; Method 2: 4 ÷ 36 = 11.1%. Method 1 is higher.

(a)(i2) (1)

B1 It uses renewable resources (water, renewable electricity) instead of a fossil fuel, and does not produce the greenhouse gas CO2.

(a)(ii1) (1)

B1 Carbon monoxide and hydrogen.

(a)(ii2) (1)

B1 $CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$

(a)(iii) (1)

B1 Chlorine (or sodium hydroxide).

(b)(i) (1)

B1 The minimum energy that colliding reactant particles must possess for a reaction to occur.

(b)(ii) (2)

B1 Curve with product level below reactant level.

B1 Ea and ΔH correctly labelled.

(b)(iii) (1)

B1 $K_c = [CH_3OH] \div ([CO][H_2]^2)$; Kc decreases. (Both items must be correct for the mark.)

(b)(iv) (3)

B1 The forward reaction is exothermic, so a higher temperature lowers the yield.

B1 Too low a temperature gives too low a rate, so 250 °C is a compromise.

B1 High pressure shifts the equilibrium position to the right, as there are fewer gas molecules on the right.

(b)(v) (1)

B1 It increases the rates of the forward and backward reactions to the same extent.

(c)(i1) (1)

B1 $4HCl(g) + O_2(g) \rightarrow 2Cl_2(g) + 2H_2O(g)$

(c)(i2) (1)

B1 $CuCl_2$ is regenerated in Step 2 and can be reused.

(c)(i3) (1)

B1 HCl is very soluble in water (it ionises completely), whereas chlorine is only slightly soluble.

(c)(ii1) (2)

B1 From Trials 1 and 2: [NO] × 2 gives rate × 4, so second order with respect to NO.

B1 From Trials 1 and 3: rate × 18 = $3^2 \times 2^{\wedge}b$, so b = 1: first order with respect to O_2.

(c)(ii2) (2)

M1 $k = 7.0 \times 10^{-6} \div (0.010^2 \times 0.010) = 7.0\ mol^{-2}\ dm^6\ s^{-1}$

A1 $\text{Rate} = 7.0 \times 0.015^2 \times 0.040 = 6.3 \times 10^{-5}\ mol\ dm^{-3}\ s^{-1}$

Section B Materials Chemistry

2. (20 marks)

(a)(i) (2)

B1 Its raw material (sugar cane) is renewable.

B1 It has the same structure as petroleum polyethene, so it is also not biodegradable.

(a)(ii) (2)

B1 Rigid rod-like shape.

B1 A polar group (C≡N) at one end, giving attraction between molecules.

(a)(iii) (1)

B1 It is a thermosetting plastic with cross-linkages, so it does not soften on heating (accept: it is an electrical insulator).

(b)(i) (1)

B1 4

(b)(ii) (2)

M1 Density = $4 \times 63.5 \div [6.02 \times 10^{23} \times (3.61 \times 10^{-8})^3]$

A1 = 8.97 g cm^{-3}

(b)(iii1) (2)

B1 Magnesium atoms differ in size from aluminium atoms and disturb the regular arrangement.

B1 Layers of atoms cannot slide over one another easily.

(b)(iii2) (1)

B1 Less energy is needed for melting and casting.

(b)(iv) (2)

M1 No. of particles is $100^3 = 10^6$ times; area of each is $1/100^2 = 10^{-4}$ times

A1 100 times

(c)(i) (1)

B1 Condensation polymerisation.

(c)(ii) (1)

B1 n HOOC–C6H4–COOH + n HOCH2CH2OH → –[CO–C6H4–CO–O–CH2CH2–O]n– + 2n H2O (equation must be balanced; accept 2n−1 H2O with end groups shown)

(c)(iii) (1)

B1 There are no cross-linkages between the polymer chains, so it softens on heating and can be reshaped.

(c)(iv) (2)

B1 PET chains contain hydrolysable ester linkages; breaking them regenerates the monomers.

B1 Polyethene has only C–C and C–H bonds, with no hydrolysable bonds.

(c)(v) (1)

B1 Extrusion moulding.

(c)(vi) (1)

B1 Other plastics would contaminate the melt and lower the quality of the recycled material.

Section C Analytical Chemistry

3. (20 marks)

(a)(i) (2)

B1 Potassium carbonate: lilac flame.

B1 Calcium carbonate: brick red flame.

(a)(ii) (1)

B1 High purity / stable in air / relatively high molar mass.

(a)(iii) (2)

B1 Iodine is non-polar, so it is more soluble in hexane than in water.

B1 Hexane and water are immiscible, so the two layers can be separated with a separating funnel.

(b)(i) (1)

B1 Purple MnO_4^- is reduced to almost colourless Mn^{2+}, so $KMnO_4$ acts as its own indicator.

(b)(ii1) (1)

B1 From colourless to pale pink (persisting).

(b)(ii2) (3)

M1 Moles of MnO_4^- = 3.32×10^{-4}; moles of H_2O_2 in 25.00 cm^3 = 8.30×10^{-4}

M1 Concentration in sample = $8.30 \times 10^{-4} \times 10 \div 0.01000 = 0.830$ M

A1 $0.830 \times 34.0 = 28.2$ g dm^{-3}

(b)(ii3) (1)

A1 $(28.22 - 30.0) \div 30.0 = -5.9\%$ (accept −6.0%), outside ±5%, so it fails.

(b)(iii) (1)

B1 MnO_4^- would oxidise Cl^- in hydrochloric acid, so the titre would be too high.

(c)(i1) (1)

A1 Empirical formula $C_4H_{10}O$ (74), and the molecular ion peak is at m/z 74, so $C_4H_{10}O$.

(c)(i2) (2)

B1 m/z 45: $CH_3CH(OH)^+$ (accept m/z 59: $C_3H_7O^+$).

B1 There is no peak at m/z 31 (CH_2OH^+), so there is no –CH_2OH group and T is not butan-1-ol. T is $CH_3CH_2CH(OH)CH_3$ (butan-2-ol).

(c)(ii) (1)

B1 The solution changes from orange to green.

(c)(iii) (2)

B1 V is $CH_3CH_2COCH_3$.

B1 T is a secondary alcohol, which is oxidised only to a ketone.

(c)(iv) (2)

B1 The spectrum of the product has no broad O–H absorption peak at 3230–3670 cm^{-1}.

B1 It shows a C=O absorption peak at 1680–1800 cm^{-1}, which T does not have.